

Notes

Notes on Elements of Set Theory

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$$B \subseteq C \implies \mathcal{P}(B) \subseteq \mathcal{P}(C)$$

Proof. By def $\mathcal{P}(B) = \{x \mid x \subseteq B\}$

$\forall x, x \in \mathcal{P}(B)$, Then, $x \subseteq B \subseteq C$

$$\implies x \subseteq C$$

$$\implies x \in \mathcal{P}(C) \text{ By def } \mathcal{P}C = \{x \mid x \subseteq C\}$$

Thus, $\mathcal{P}(B) \subseteq \mathcal{P}(C)$

□

$$A \text{ is a finite set, } |A| = n \implies |\mathcal{P}(A)| = 2^n$$

Proof. Consider taking i elements from A to build a subset.

Then $i \in \{0, \dots, n\}$, and the number of that kind of subsets is $\binom{n}{i}$.

(if $i = 0$, the subset is $\emptyset$; if $i = n$, the subset is A itself)

$$\implies |\mathcal{P}(A)| = \sum_{i=0}^n \binom{n}{i}$$

$$\implies |\mathcal{P}(A)| = 2^n, \text{ By Binomial Theorem, consider } (1 + 1)^n$$

□

$$x, y \in B \implies \{\{x\}, \{x, y\}\} \in \mathcal{P}(\mathcal{P}(B))$$

Proof. $x, y \in B$

$$\implies \{x\}, \{x, y\} \in \mathcal{P}(B)$$

$$\implies \{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(B)$$

Thus, by def $\{\{x\}, \{x, y\}\} \in \mathcal{P}(\mathcal{P}(B))$

□

Remark. A simple way to calculate the rank of set (which finite) is to count the deepest nested brackets until meet atom or $\emptyset$.

$$\forall \alpha \in \mathbb{N}, \alpha \leq 3, V_{\alpha+1} = A \cup \mathcal{P}(V_\alpha)$$

Proof. $V_0 = A$

$$\text{By def, } V_{n+1} = V_n \cup \mathcal{P}(V_n)$$

$$V_1 = V_0 \cup \mathcal{P}(V_0) = A \cup \mathcal{P}(V_0)$$

$$V_2 = V_1 \cup \mathcal{P}(V_1) = A \cup \mathcal{P}(V_0) \cup \mathcal{P}(V_1) = A \cup \mathcal{P}(V_1)$$

$\mathcal{P}(V_0) \cup \mathcal{P}(V_1) = \mathcal{P}(V_1)$, By $\mathcal{P}(V_0) \subseteq \mathcal{P}(V_1)$ since $V_0 \subseteq V_1$, by previous result.

$$V_3 = V_2 \cup \mathcal{P}(V_2) = A \cup \mathcal{P}(V_1) \cup \mathcal{P}(V_2) \text{ Similarly,}$$

$$V_3 = A \cup \mathcal{P}(V_2)$$

□

Remark. I guess we can use mathematical induction to make it to $\forall \alpha \in \mathbb{N}$

Remark. All sets are classes, $\{x \mid \phi(x)\}$ is appropriately is class-builder, where $\phi(x)$ is open formula.

Remark. Definition is like an abbreviation of syntax,
and it seems that it must be iff statements.

Remark. $\forall x(x \in B \iff \phi(x)) \iff B = \{x \mid \phi(x)\}$

Axiom 1. Extensionality Axiom

$$\forall A \forall B, (\forall x, x \in A \iff x \in B) \implies A = B$$

Axiom 2. Empty Set Axiom

$$\exists A \forall x, x \notin A$$

Remark. Does $\emptyset$ act like a unique generator in terms of abstract algebra?
(e.g. similar as prime numbers can be used to generate whole $\mathbb{N}$ under multiplication)

Axiom 3. *Pairing Axiom*

$$\forall A \forall B \exists C \forall x (x \in C \iff x = A \vee x = B)$$

Axiom 4. *Union Axiom*

Axiom 5. *Power Set Axiom*

$$\forall A \exists B \forall x (x \in B \iff x \subseteq A)$$

Axiom 6. *Subset Axiom*

$$\forall t_1 \dots \forall t_n \forall c \exists B \forall x (x \in B \iff x \in c \wedge \phi(x, t_1, \dots, t_n))$$

Remark. It is an axiom schema, where ϕ is a first-order formula;
The number of subset axioms would be $\aleph_0$;
 $P(x)$ (predicate) includes $\phi(x)$ (open formula) and σ (closed formula).

Axiom 7. *Replacement Axiom*

Axiom 8. *Infinity Axiom*

Axiom 9. *Regularity Axiom*

Axiom 10. *Choice Axiom*